

Rahul Natesan

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EDUCATION

- Vellore Institute of Technology (VIT)** Vellore, India
• *Bachelor of Technology - Computer Science & Engineering; CGPA: 8.86/10* July 2023 - August 2027
Courses: Data Structures & Algorithms, OOP, Design Patterns, Operating Systems, Computer Networks, Database Management Systems, Artificial Intelligence, Cloud Architecture

SKILLS SUMMARY

- **Languages:** Java (Core), Python, C++, JavaScript, SQL
- **Frameworks:** React.js, Node.js, Express.js, Spring Boot, FastAPI, REST APIs, Redux
- **Cloud & DevOps:** AWS Lambda, Amazon SQS, EventBridge, API Gateway, S3, RDS, Docker, Git, CI/CD (GitHub Actions)
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Machine Learning:** Scikit-learn, TensorFlow, Pandas, NumPy
- **Soft Skills:** Communication, Vendor Coordination, Cross-functional Collaboration, Event Management

EXPERIENCE

- Apexon (Infostretch Corporation India Pvt. Ltd.)** Hyderabad, India
• *Software Engineering Intern* May 2026 - Jul 2026
 - **Serverless Architecture:** Architected and deployed a cloud-native real-time stock market analytics system on AWS using 6 Lambda functions, Amazon SQS, EventBridge, API Gateway WebSockets, RDS MySQL, and S3, processing live NSE data every minute across 9 stocks.
 - **Event-Driven Pipeline:** Engineered a fully serverless ingestion-to-dashboard pipeline generating 7,000+ database operations/day with real-time MA5, MA15, percentage change, and candlestick trend analytics for live stock monitoring.
 - **React Dashboard:** Developed a WebSocket broadcasting layer with sub-second delivery latency and deployed a React + TypeScript + Tailwind CSS dashboard on S3 with live KPI visualization and top movers tracking.
- KuppiSmart Solutions** Remote
• *Software Developer Intern* May 2025 - Jul 2025
 - **Frontend & Component Library:** Built 10+ responsive pages and a reusable component library using React.js and Tailwind CSS; reduced layout shift by 30% through optimized render paths and lazy loading, improving page load performance.
 - **Payment Integration:** Integrated Razorpay payment gateway via secure REST APIs serving 500+ transactions; monitored conversion analytics over 2-week agile sprints and delivered 3 iterative feature releases.
 - **CI/CD Automation:** Automated build, test, and deploy pipeline using GitHub Actions; cut manual release time by 40% and enforced code-style guidelines across a 5-member engineering team.

PROJECTS

- **Real-Time Stock Market Analytics Platform (AWS, Serverless, React, TypeScript):** Architected a cloud-native analytics system using 6 AWS Lambda functions, Amazon SQS, EventBridge, API Gateway WebSockets, RDS MySQL, and S3; engineered an event-driven pipeline processing 3,500+ NSE stock records/day with real-time MA5, MA15, and candlestick analytics. Tech: Python, React, TypeScript, Tailwind CSS, AWS (Jun '26)
- **AI Disease Prediction System (Machine Learning, Flask, REST API):** Designed a symptom-based diagnostic system using supervised Machine Learning (ExtraTrees, Naive Bayes, KNN, Neural Networks); achieved 95% accuracy through feature selection and hyperparameter tuning. Deployed as a Flask REST API with unit tests validating model reliability. Tech: Python, Flask, Scikit-learn, TensorFlow, Pandas (Sep '24)
- **AgriConnect Platform (Java Spring Boot, React.js, Microservices):** Built scalable RESTful microservices backend using Java Spring Boot; applied OOAD and Design Patterns for enterprise-grade maintainability. Integrated real-time data feeds and chatbot support improving farmer decision-making outcomes. Tech: Java Spring Boot, React.js, Tailwind CSS, MySQL (Apr '24)

CERTIFICATIONS

- **AI Agents & Agentic AI:** Vanderbilt University via Coursera — July 2025
- **Python Essentials 1 & 2:** Cisco Networking Academy — July 2025
- **Data Science:** Teachnook in association with IIT Roorkee — June 2024

ACTIVITIES & LEADERSHIP

- **Riviera — VIT International Cultural Fest, Volunteer:** Food Stall Committee; coordinated with 10+ vendors, managed stall logistics and real-time issue resolution. Feb 2024
- **Gravitas — VIT Technical Fest, Coordinator:** Food Stall Committee; led vendor onboarding and engagement, demonstrating communication and stakeholder management. Feb 2025
- **Industry Academia Conclave — 24-Hour Hackathon, Winner:** Led end-to-end solution design, rapid prototyping, and judge-facing demo; demonstrated ownership and delivery under pressure. Sept 2025